

VIRGEN, Austria

WMO: 112520

Lat: 47.003N

Lon: 12.456E

Elev: 1193

StdP: 87.79

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-10.2	-17.9	0.9	-10.5	-15.9	1.1	-8.1	4.4	-1.6	3.8	-1.5	1.1	270	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	26.4	16.7	24.6	16.0	22.8	15.3	17.8	24.0	16.9	22.6	16.1	21.3	2.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	13.1	19.3	15.0	12.3	18.4	14.2	11.7	17.9	55.1	24.2	52.3	22.8	49.7	21.3	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.0	4.1	3.4	-13.4	29.1	2.7	1.8	-15.4	30.4	-17.0	31.4	-18.6	32.4	-20.6	33.7		
			-14.4	19.2	2.6	1.0	-16.3	19.9	-17.8	20.5	-19.3	21.1	-21.2	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-2.4	-0.8	2.9	6.7	10.9	14.5	16.3	15.7	11.7	7.8	2.6	-1.0
	DBStd	7.48	3.90	4.28	4.24	3.82	3.40	3.68	3.10	3.29	3.19	3.53	3.86	4.09
	HDD10.0	1721	384	301	220	110	30	6	1	2	19	84	224	341
	HDD18.3	4124	642	534	477	349	231	123	78	93	199	326	473	599
	CDD10.0	672	0	0	1	11	58	141	195	179	69	16	1	0
	CDD18.3	35	0	0	0	0	1	8	14	13	0	0	0	0
	CDH23.3	300	0	0	0	0	9	79	121	88	2	0	0	0
	CDH26.7	38	0	0	0	0	0	9	16	13	0	0	0	0
Wind	WSAvg	1.3	1.1	1.2	1.4	1.6	1.6	1.5	1.4	1.3	1.1	1.1	1.0	1.0
Precipitation	PrecAvg	1038	43	36	54	63	101	136	154	149	99	91	70	44
	PrecMax	1294	87	93	164	127	145	191	222	223	153	181	164	107
	PrecMin	881	10	3	13	3	41	78	55	95	47	8	0	0
	PrecStd	100	22	23	33	33	32	32	37	34	28	44	39	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	12.3	16.2	20.0	24.8	27.8	28.8	28.5	23.2	18.4	13.7	10.6
		MCWB	4.1	6.1	8.3	11.7	14.8	16.6	17.8	17.9	15.6	12.5	8.2	4.9
	2%	DB	6.2	9.2	13.8	17.3	21.6	25.7	26.2	25.9	20.7	16.3	10.7	7.1
		MCWB	2.4	4.1	7.2	9.9	13.4	16.1	16.7	17.0	14.3	11.3	6.8	2.8
	5%	DB	4.3	6.7	11.5	15.3	19.5	23.6	24.5	23.6	18.8	14.5	8.8	5.1
		MCWB	1.3	2.5	5.9	8.8	12.2	15.2	16.1	16.2	13.5	10.5	6.0	2.1
	10%	DB	2.6	4.7	9.2	13.2	17.3	21.5	22.5	21.7	17.0	12.7	7.3	3.7
		MCWB	0.1	1.4	4.5	7.7	11.1	14.2	15.2	15.4	12.7	9.7	5.3	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	6.4	9.1	12.2	15.6	18.1	18.8	19.1	16.5	13.6	9.2	5.4
		MCDB	7.6	11.4	15.2	19.4	22.7	24.0	26.0	25.1	21.0	16.7	11.8	8.8
	2%	WB	2.9	4.4	7.6	10.5	14.1	16.8	17.7	17.9	15.3	12.1	7.7	3.8
		MCDB	5.2	8.6	12.9	16.3	20.3	23.8	23.9	24.1	19.0	15.1	9.5	6.0
	5%	WB	1.7	2.9	6.3	9.2	13.0	15.9	16.9	16.9	14.2	11.0	6.7	2.7
		MCDB	3.7	6.0	10.9	14.4	18.3	22.0	22.5	22.2	17.5	13.6	8.2	4.5
	10%	WB	0.6	1.7	5.0	8.1	11.7	14.9	16.1	16.0	13.3	10.1	5.6	1.6
		MCDB	2.2	4.2	8.6	12.4	16.1	20.3	21.0	20.6	16.2	12.4	7.1	3.3
Mean Daily Temperature Range	5% DB	MDBR	5.8	7.2	8.7	9.0	9.2	9.6	9.8	9.2	8.5	7.4	5.6	5.4
		MCDBR	7.5	9.9	11.9	12.3	12.8	13.3	12.9	12.3	11.0	9.5	7.5	6.8
		MCWBR	5.5	6.7	7.1	6.5	6.1	5.5	5.2	5.2	5.9	5.8	5.3	5.0
	5% WB	MCDBR	6.6	9.0	11.2	11.5	11.8	12.0	11.2	11.1	9.4	8.5	6.5	6.2
		MCWBR	5.1	6.3	6.9	6.3	5.8	5.3	5.2	5.2	5.4	5.5	4.8	4.8
Clear-Sky Solar Irradiance	taub	0.254	0.280	0.326	0.371	0.386	0.395	0.391	0.385	0.350	0.324	0.290	0.256	
	taud	2.353	2.271	2.237	2.237	2.271	2.290	2.303	2.339	2.413	2.457	2.479	2.425	
	Ebn at Noon	851	890	893	881	879	869	868	859	860	832	802	810	
	Edh at Noon	70	95	116	129	129	128	125	115	98	80	64	59	
All-Sky Solar Radiation	RadAvg	1.39	2.29	3.34	4.18	4.82	5.38	5.48	4.83	3.58	2.44	1.40	1.16	
	RadStd	0.16	0.30	0.40	0.54	0.58	0.49	0.47	0.52	0.44	0.28	0.25	0.17	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (87 neighbors)	+0.43	N/A	N/A	N/A	+0.41	+0.38	N/A	-119	+73	+30	

Nomenclature: See separate page